

Metrics of Left Ventricular Active Relaxation Reflect Proteomic Myocardial Remodelling and Reverse Remodelling

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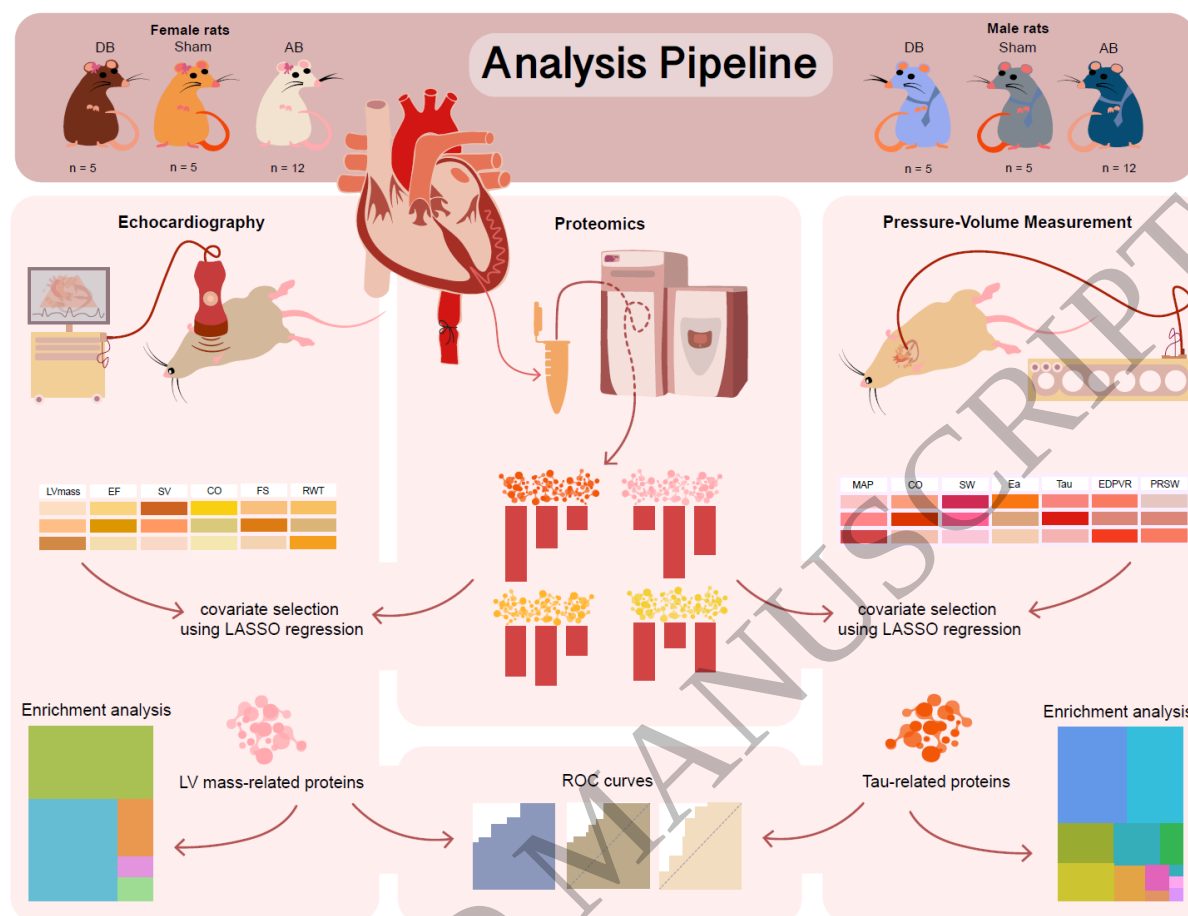
Abstract

Aims: Pressure overload (PO) induces adverse myocardial remodelling, whereas efficient pressure unloading interventions can promote reverse remodelling at structural and molecular levels. This study seeks to identify a metric of left ventricular (LV) performance that reflects the degree of myocardial remodelling and reverse remodelling.

Methods: Male and female rats underwent surgical aortic constriction/banding (AB) to generate pressure overload (PO) and subsequent debanding (DB) to induce pressure unloading. The architecture and function of the left ventricle were evaluated using echocardiography and pressure-volume analysis. Exploratory proteome profiling via LC-MS/MS on left ventricular samples was followed by LASSO-based feature selection on existing data to determine LV parameters linked to myocardial proteomic changes.

Results: Tau (τ), the time constant of LV isovolumic pressure decay and an index of active relaxation, and LV mass showed the strongest associations with proteome-wide abundance changes. With 842 associated proteins and 154 overrepresented gene ontology terms, Tau demonstrated a strong correlation with LV proteomic changes during myocardial remodelling, and reverse remodelling. Tau-associated proteins had an AUC > 0.85 when discriminating remodelled proteomes of AB rats from non-banded counterparts. Network analysis reveals varied protein-protein correlations across circumstances and indicates transcription factors regulating Tau-associated proteome changes. Across sham (Co), 6- and 12-week banding (AB6/AB12), and debanding after 6 weeks followed by 6 weeks unloading (DB), τ showed the largest set of associated proteins (n=842) and enriched processes.

Conclusions: LV active relaxation, quantified by Tau, most comprehensively reflects myocardial proteomic remodelling during pressure overload and unloading. Indices of diastolic relaxation may therefore represent integrative functional readouts of molecular remodelling and potential translational markers of reverse remodelling.



Graphical Abstract. Illustration of data processing steps, including data collection and feature selection. AB = aortic banded, DB = debanded, EF = ejection fraction, SV = stroke volume, CO = cardiac output, FS = fractional shortening, RWT = relative wall thickness, MAP = mean arterial pressure, SW = stroke work, Ea = arterial elastance, EDPVR = end-diastolic pressure-volume relationship, PRSW = preload-related stroke work.

Introduction

Pathological, or adverse, myocardial remodelling refers to changes in ventricular size, shape, and geometry due to pressure overload (induced by hypertension or aortic stenosis), volume overload (induced by valvular regurgitation), and inflammatory diseases¹. These changes can negatively impact myocardial function, with pressure and volume overload causing ventricular systolic function to deteriorate, and inflammation inducing structural changes in the myocardium¹. These alterations include molecular, cellular, metabolic and biochemical changes, as well as inflammatory response, and altered protein synthesis².

While adverse myocardial **remodelling** is an important cause of progressive myocardial dysfunction and leads to a poor prognosis in heart failure and clinical outcomes, myocardial reverse **remodelling** refers to a favorable remission of the myocardium in terms of geometry, morphology, and function^{3,4}. Reverse

remodelling involves a cascade of changes in cellular and metabolic regulatory mechanisms and may be caused by pressure and volume unloading⁵. As its induction by pharmacological, non-invasive, or surgical techniques has been shown to decrease mortality and improve the effects of heart failure, reverse **remodelling** is thereby of high relevance⁶. To acquire further insights into the molecular mechanisms of such treatment strategies, proteomic analyses can be conducted.

Existing approaches to assess reverse **remodelling** primarily focus on evaluating changes in left ventricular (LV) end-systolic or end-diastolic volume, often expressed as relative or absolute reductions thereof^{7,8}. However, these approaches do not capture the molecular mechanisms underlying reverse **remodelling** and current therapies for promoting reverse **remodelling**. In contrast, proteomics, as used in the scope of this study, could help in determining unifying molecular and functional parameters to quantify reverse **remodelling** and to predict clinical interventional effectiveness^{9,10}.

Methods

Proteomics dataset and animal model

In this study, we extensively reanalyze the proteomics data from our previously published study¹¹, which are made available on the MassIVE repository (MSV000089077). The dataset involves data on Sprague-Dawley rats treated per the EU Directive 2010/63/EU and the National Institutes of Health's *Guide for the Care and Use of Laboratory Animals* (NIH Publication No. 85-23, Revised 1996), and according to the ARRIVE guidelines¹². The regional authorities for Animal Experimentation in Karlsruhe, Germany approved this investigation (G-198/16). The details of care for and use of the animals have been described previously¹¹. The cohort included several experimental groups: a sham-operated group as control (n = 10, Co), aortic banded groups observed for 6 or 12 weeks (n = 12 each, AB) and a debanded group (n = 10, DB) that underwent the AB operation, was observed for 6 weeks, then underwent debanding after which they were followed up until week 12. Each group included an equal number of male and female rats. Aortic banding (the surgical procedure of constricting the abdominal aorta of the rats to the external diameter of a 22-gauge needle between the superior mesenteric artery and the right renal artery, described in detail elsewhere¹¹) leads to pressure overload and was used for modelling myocardial remodelling and hypertrophy. The reversal of the constriction, debanding, leads to pressure unloading and was used for modelling myocardial reverse remodelling. **Terminal echocardiographic and PV assessments were performed at the end of the respective observation periods (week 6 for AB6; week 12 for AB12, and DB). Debanded animals underwent 6 weeks of banding followed by surgical removal of the constriction and a further 6-week unloading period prior to terminal analysis.**

Assessment of left-ventricular morphology and function.

Echocardiography measurements are described in detail elsewhere¹¹. In brief, the Vevo 2100 imaging system (FujiFilm VisualSonics, Inc., Toronto, ON, Canada) equipped with a 21-MHz linear probe was utilized to obtain the sonographic measurements at the end of the observation period. Images were recorded in two-dimensional parasternal long- and short-axis views, as well as in M-mode at the midpapillary level, followed by the analysis of the digital images in a blinded fashion. Based on the measurements relative wall thickness (RWT), fractional shortening (FS), ejection fraction (EF), stroke volume (SV), cardiac output

(CO), and left ventricular mass (LV mass) were calculated. LV pressure-volume (P-V) analysis with a 2F microtip pressure-conductance catheter (SPR-838, Millar Instruments, Houston, TX), inserted into the right carotid artery and advanced into the ascending aorta and then into the left ventricle, was performed to assess various aspects of LV function in steady state and during the transient occlusion of the inferior vena cava: mean arterial pressure (MAP), cardiac output (CO), ejection fraction (EF), arterial elastance (E_a ; calculated by the following equation $E_a = \text{LVESP}/\text{SV}$)¹³, and time constant of LV pressure decay (τ ; according to the Glanz method)¹⁴, stroke work (SW), preload recruitable stroke work (PRSW), slope of the end-diastolic pressure-volume relationship (EDPVR). Parameters were calculated using a special P-V analysis program (PVAN, Millar instruments).

Reanalysis of proteomics data

The FragPipe pipeline 19–21 (v17) was used for peptide/protein identification and quantitation¹⁵. In brief, spectral data was searched against an in silico-digested protein sequence database (EBI Rat canonical proteome, version 2021_03 appended with common contaminants and iRT peptides), assuming experimental tryptic digestion to generate in silico-digested peptides and precursor candidates with a mass tolerance of -20/20 ppm. Peptide N-terminal acetylation and peptide N-terminal TMT labelling were set as variable modifications. MSBooster was used for deep learning-based predictions of retention time and spectra. Predicted features were used by Percolator 4 for post-processing scoring and false discovery rate (FDR) control via target-decoy competition of all peptide-to-spectrum matches (PSMs) obtained from MSFragger search. Relative quantitation of identified peptides within each sample was performed via their reported ion intensities using TMT-Integrator. Only PSMs derived from unique peptides with a minimum probability of 0.9 and an isotopic purity of at least 50% were considered for quantitation. Quantitative values were normalized via median centering and summarized by protein/gene and peptide using a virtual reference channel to finally generate protein and peptide normalized matrices of abundance, which then underwent statistical processing.

Statistics

Feature selection of the clinical parameters was performed using Least Absolute Shrinkage and Selection Operator (LASSO) regression to identify the minimum fset of associations of protein abundances with LV parameters. The best lambda, i.e., the best regularization parameter controlling the strength of the penalty applied to regression coefficients, was determined following the application of LASSO to all predictors by k-fold cross-validation (with k=10) as implemented in the cv.glmnet function of the glmnet package¹⁶. Subsequently, for all proteins with non-zero coefficients, the Linear Models for Microarray Data (limma)¹⁷ was run on PV and echocardiographic parameters obtained by LASSO to estimate coefficients without regularization. A protein was deemed significantly associated with a PV or echocardiographic parameter if the limma p-value was less than 0.05. The highest-performing PV and echocardiographic parameters were then selected for subsequent investigation. The term “highest-performing” denotes the pressure-volume and echocardiographic parameters with the largest number of associated proteins. Gene Ontology (GO) enrichment analysis (GO over-representation analysis), facilitated by clusterProfiler¹⁸, was conducted on proteins associated with selected parameters of the left ventricle. To offer a comprehensive overview of enriched biological processes, GO: BP terms were grouped hierarchically based on semantic similarity (rrvgo)¹⁹. Sensitivity and specificity of Tau and LVmass-associated proteins was calculated across different thresholds to identify the best biomarker candidates, the relative abundance of which can best discriminate

non-banded (either sham-operated or debanded) animals from ongoing myocardial remodelling (AB). Proteins with an AUC of more than 0.85 on the ROC curves were considered good discriminators. **Exclusively for the ROC analyses, sham-operated controls (Co) and debanded animals (DB) were pooled as a “non-banded” reference group. Both groups represent the absence of ongoing pressure overload at terminal sampling (Co: never banded; DB: pressure unloading following prior banding). In the original cohort, Co and DB demonstrated high functional and proteomic similarity at week 12.**

Network analysis

The online Search Tool for the Retrieval of Interacting Genes/Proteins online database (STRING-DB) tool²⁰ was used to create a protein-protein interaction (PPI) network of all Tau-related proteins. The visualization allows the investigation of potential interactions among Tau linked proteins²¹. Subsequently, the MCODE plugin²² of the Cytoscape software²³ was used for subnetwork extraction and re-identifying densely connected regions/modules associated with Tau from the STRING network with the following settings: node score cut-off ≥ 0.2 , degree cut-off ≥ 2 , max depth = 100 and K-core ≥ 2 . Subnetworks with an MCODE score > 3 were selected for further evaluation. Pairwise relationship among subnetwork member proteins was investigated with Pearson correlation separately, in the different conditions (Co, AB, DB).

MCODE Subnetwork proteins subsequently underwent correlational analysis with AB, DB and Control (Co) and plotted in separate network plots based on edge-weighted spring-relationship, where a spring model is used to simulate relative attraction between proteins (nodes).

Both the STRING Database combined score for inter-node relationships for Tau-related proteins and correlation coefficients for subnetwork proteins were used as determinants for graphically representing subnetwork proteins, node positioning and edge weight representing the edge-weighted spring-relationship. Proximity and edge weight in subnetworks were selected to be the graphical markers of increased correlation and combined score.

Correlation between inter-group proteins was used as the edge weight determinant, with shorter distances indicating greater correlation. The correlation of AB, DB and control proteins with other proteins in each subnetwork was indicated using edge width between individual nodes. Nodes and edges were color-coded to enable visualization of those proteins identified as being related to cardiac function and / or had AUC of more than 0.85 (banded vs. non-banded rats).

Transcription factors were identified using the ChEA3 algorithm²⁴ based on the ENCODE database²⁵ to export the 10 transcription factors with the lowest FDR-corrected p-values (< 0.05) and their potential Tau-associated target proteins.

Results

Functional and structural characteristics across experimental groups

As expected, aortic banding induced a stepwise phenotype beginning with increased afterload, as MAP and Ea were both elevated in AB6 and AB12, whereas debanding reduced MAP toward control values and attenuated Ea, particularly relative to AB12. These hemodynamic changes were accompanied by concentric remodeling, reflected by increased RWT and LV mass in the banded groups, with partial regression after debanding (Table 1 & 2). Systolic impairment was more modest, with EF and FS tending to decrease and

EF being significantly reduced in AB12, while PRSW was increased in AB6 and normalized in DB. Diastolic dysfunction was most clearly captured by Tau and, to a lesser extent, EDPVR: both were higher in the banded groups, and after debanding EDPVR returned to control-like values while Tau improved markedly toward Co (Table 2).

Echocardiography

Parameter	Co (n=10)	AB6 (n=12)	AB12 (n=12)	DB (n=11)
RWT (-)	0.48 ± 0.02	0.57 ± 0.02	0.60 ± 0.05	0.52 ± 0.03
FS (%)	43.69 ± 1.19	38.81 ± 1.53	37.03 ± 2.35	40.21 ± 2.20
EF (%)	56.87 ± 1.65	53.98 ± 2.32	47.23 ± 2.14*	54.47 ± 2.68
SV (uL)	238.01 ± 11.07	232.08 ± 12.23	231.41 ± 20.13	225.35 ± 13.57
CO (mL/min)	84.44 ± 6.30	80.81 ± 4.49	75.22 ± 6.94	79.01 ± 4.51
LV mass (mg)	1219 ± 107	1563 ± 105	1708 ± 145*	1291 ± 118

Values are presented as mean ± SEM; * p < 0.05 vs Co, # DB p < 0.05 vs AB6, \$ DB p < 0.05 vs AB12.

Table 1. Left ventricular structure and systolic function measured by echocardiography across control, pressure-overload (AB6, AB12), and debanded (DB) experimental groups.

Pressure-volume analysis

Parameter	Co (n=10)	AB6 (n=12)	AB12 (n=12)	DB (n=11)
CO (uL/min)	59564 ± 3115	66440 ± 3720	57341 ± 3790	61106 ± 4088
EDPVR (mmHg/uL)	0.03 ± 0.003	0.05 ± 0.01	0.05 ± 0.005	0.03 ± 0.002
PRSW (mmHg)	104.66 ± 10.27	170.63 ± 14.54*	132.67 ± 11.34	103.64 ± 8.88#
SW (mmHg*uL)	16648 ± 1120	24314 ± 1553*	22429 ± 1944	19894 ± 1437
Ea (mmHg/uL)	0.80 ± 0.04	1.15 ± 0.09*	1.44 ± 0.10*	0.95 ± 0.09\$
MAP (mmHg)	122.51 ± 3.01	179.29 ± 5.03*	174.17 ± 5.56*	129.51 ± 4.87#\$
Tau (msec)	12.72 ± 0.33	16.01 ± 0.69*	17.62 ± 1.04*	13.64 ± 0.76\$

Values are presented as mean ± SEM; * p < 0.05 vs Co, # DB p < 0.05 vs AB6, \$ DB p < 0.05 vs AB12.

Table 2. Left ventricular hemodynamic parameters obtained by pressure–volume analysis in control, pressure-overload (AB6, AB12), and debanded (DB) groups.

Selecting parameters of pressure-volume analysis and echocardiography most reflective of myocardial proteomic alterations

Covariate selection using LASSO regression for the using either the PV parameters or the echocardiographic parameters as predictors for protein abundances, we found one parameter per modality that clearly outperformed the others. The highest number of proteins with a non-zero coefficient and significant limma p-value were obtained for Tau (the time constant of active left ventricular relaxation) as determined by PV analysis and for LV mass as measured using echocardiography during myocardial remodelling and reverse remodelling (Figure 1). Thus, the following analyses are focused on these two key ventricular parameters to uncover their associated proteomic alterations during myocardial remodelling.

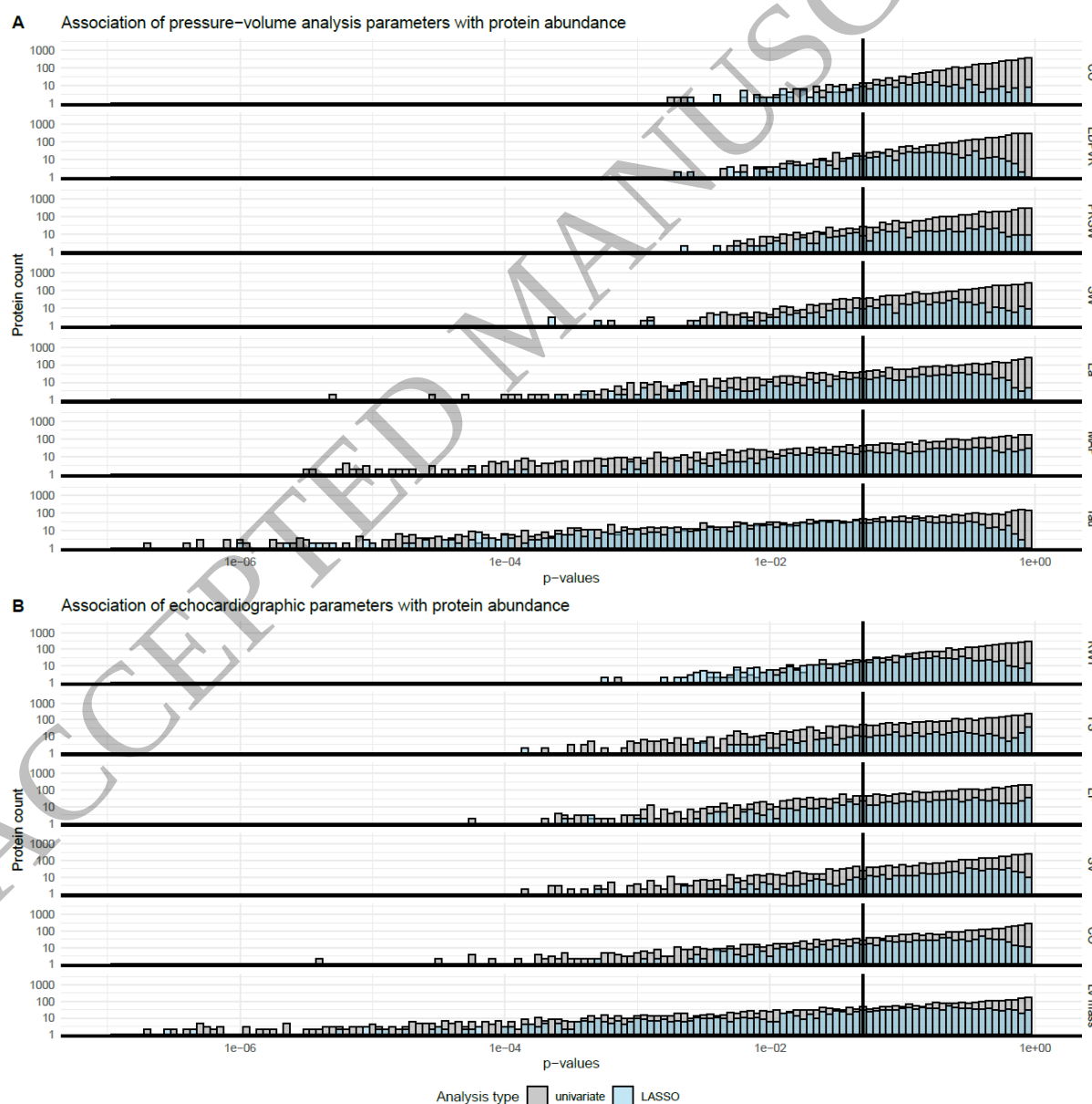


Figure 1. Number of proteins associated with sensitive parameters of PV analysis and echocardiography; determined through univariate analysis (limma conducted on each parameter separately, p -value < 0.05) of all proteins and multivariate analysis of all proteins with feature selection by LASSO regression. For LASSO, we performed a subsequent limma analysis and define parameters with p -value < 0.05 as significantly associated with PV or echocardiographic parameters. Panel A illustrates that Tau emerges as the PV parameter associated with the highest number of proteins, and Panel B highlights LV mass as the most predominant echocardiographic parameter.

Tau surpasses LV mass in both diversity and quantity of protein associations

Gene Ontology enrichment analysis allowed for the identification of significant biological processes, molecular functions, and cellular components associated with both Tau and LV mass-linked proteins. Notably, Tau, being associated with abundance changes of 842 proteins, displayed a more comprehensive enrichment profile compared to LV mass, which was associated with 568 proteins. In terms of cellular components, Tau shows a stronger association with the protein expression of structural components of the sarcomere ('supramolecular complex', 'myofibril', 'contractile fiber', 'I band') and the spliceosomal complex. Enrichment of molecular functions results in various gene ontology terms related to DNA binding, regulation of transcription, as associated with alterations in Tau and LV mass during myocardial remodelling and reverse remodelling (Figure 2).

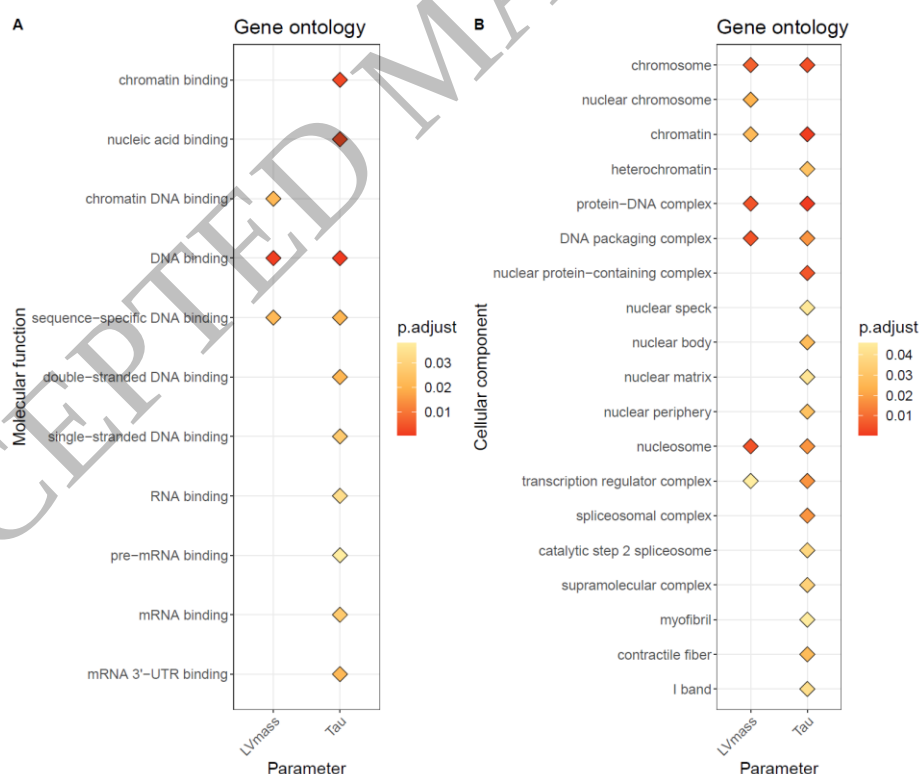


Figure 2. Gene Ontology (over-representation) enrichment analysis for Tau and LV mass-related proteins, identifying the molecular functions and cellular compartments of proteins associated with each parameter. The continuous color scale illustrates respective FDR adjusted p -values.

1 Proteins related to Tau are connected to diverse biological processes. Notably, of the 842 Tau-related
2 proteins, 47 are connected to cardiac muscle cell development (Figure 3). Further, we were able to identify
3 that Tau-related proteins are involved in gene expression regulation and mRNA splicing within the
4 spliceosome machinery. Additionally, Tau-related proteins are connected to functions like single-stranded
5 DNA binding, RNA binding, and transcriptional coregulation and coactivation.

6 Conversely, LV mass was linked to a more confined set of 568 proteins. Those indicate different aspects of
7 myocardial adaptation, with enriched functions of these proteins including DNA-binding transcription
8 factor binding and protein heterodimerization activity (Figure 3). Despite fewer proteins being associated
9 with LV mass than with Tau, Gene Ontology terms, such as chromatin binding, nucleic acid binding, DNA
10 binding, and transcription factor binding, were shared among Tau and LV mass.

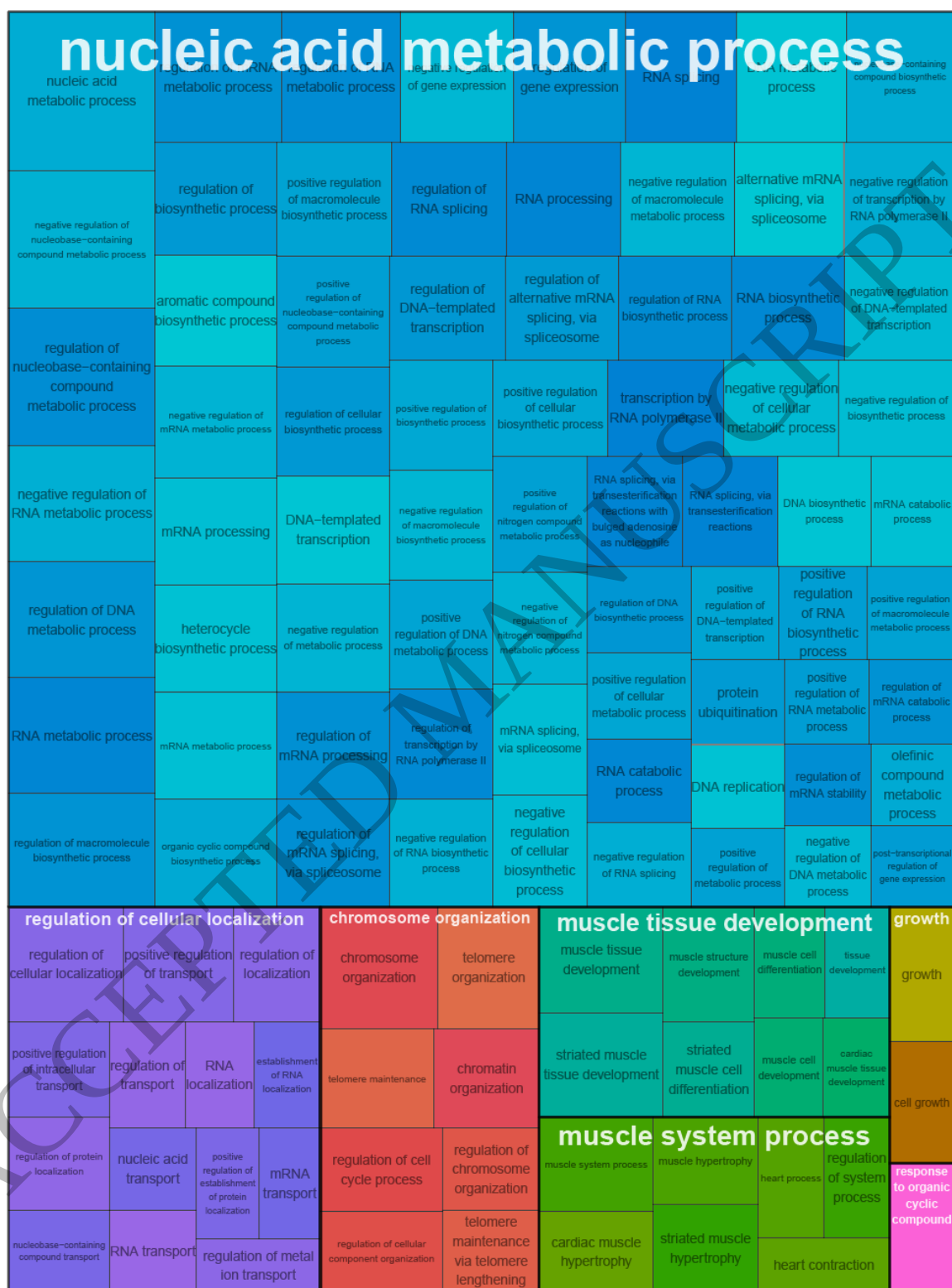


Figure 3. Enrichment of biological process for Tau-related proteins. To provide a better overview, terms were automatically grouped based on semantic similarity.

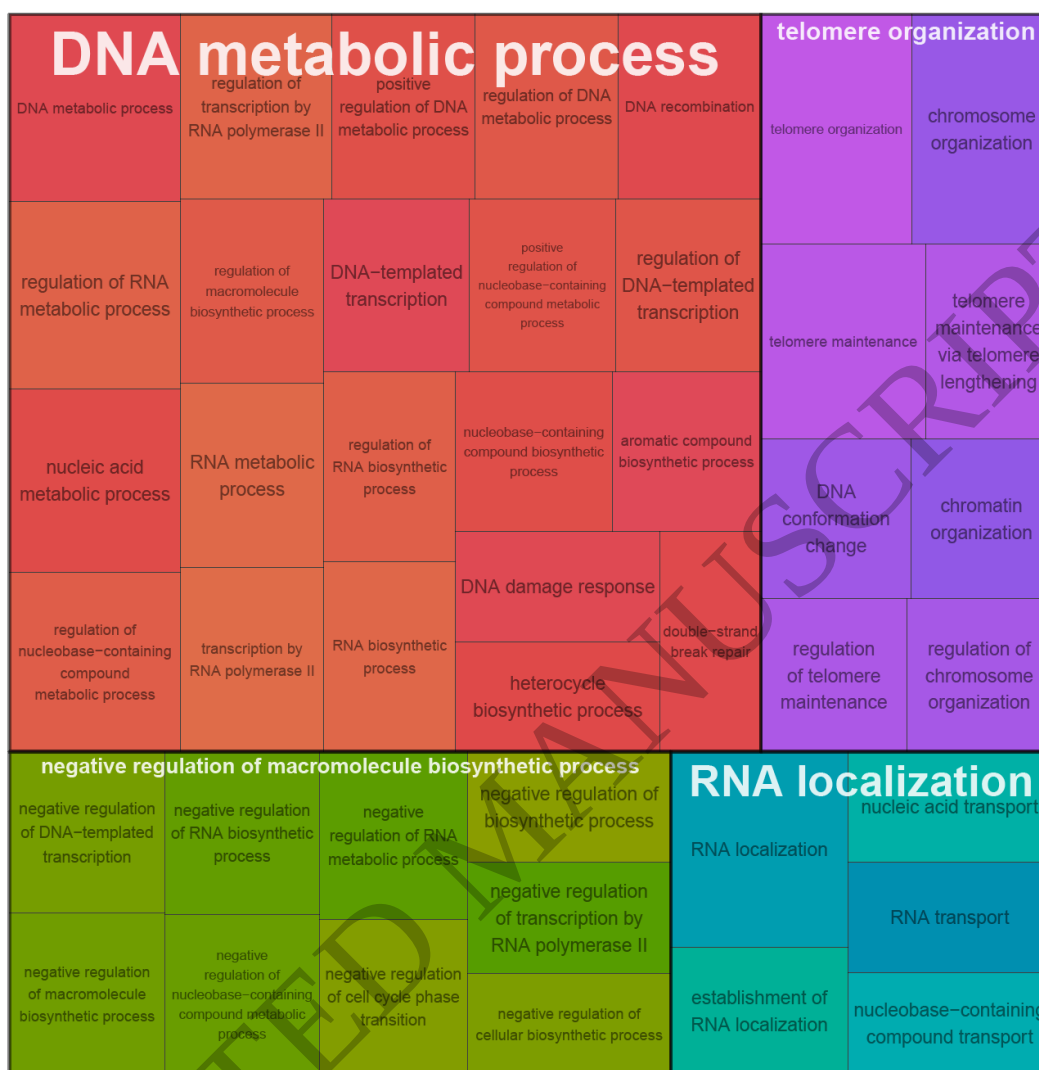


Figure 4. Enrichment of biological processes for LV mass-related proteins. To provide a better overview, terms were automatically grouped based on semantic similarity.

Several Tau-related proteins can be used to detect active myocardial remodelling with high accuracy

To investigate the potential of our proteins of interest being used as biomarkers for myocardial remodelling, Co and DB animals were combined into a ‘non-banded’ group, based on the high functional and proteomic similarity of these conditions, substantiated by our previous findings in the same cohort¹¹. Non-banded animals, thus, represent inactive remodelling, while AB groups active myocardial remodelling. Receiver operating characteristic (ROC) curves of (Figure 5) the proteins associated with Tau and/or LV mass revealed a subset of proteins that exhibited an Area under the curve (AUC) higher than 0.85, effectively differentiating the active remodelling of aortic banded (AB) rats from non-banded conditions. Of these, 19 Tau-related proteins were identified, compared to only 9 proteins associated with LV mass. Proteins identified as being related exclusively to Tau included *ANXA5*, *GSTK1*, *COQ9*, *TMOD4*, *PCCA*, *PDLIM5*, *FBLIM1*, *EIF4G1*, *SORBS2*, *TST*, *PSME1*, *DBT*, and *NRAP*.

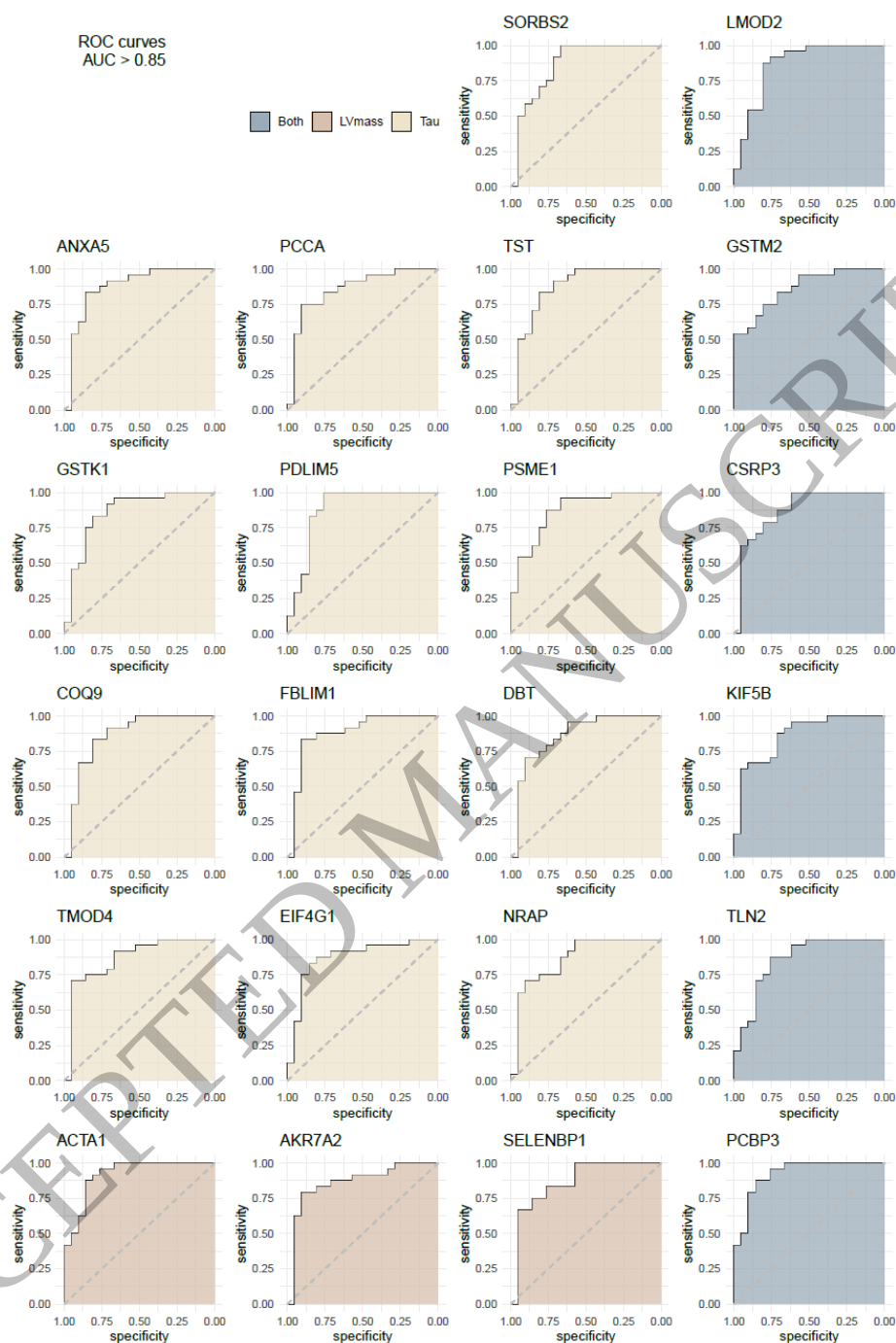


Figure 5. ROC curves of proteins associated to Tau with an AUC of more than 0.85, based on the calculation of sensitivity and specificity across different thresholds, differentiating active myocardial remodelling (AB) from non-banded (CO, DB) rat proteomes.

Molecular Complex Detection analysis reveals a network of closely interconnected proteins

To investigate the potential interplay among Tau-associated proteins in myocardial remodelling and reverse remodelling, protein-protein interaction networks were acquired from the STRING database. Based on the

subset of Tau-associated proteins a network of 839 closely interconnected nodes with 6142 edges was detected using the MCODE plugin²² of Cytoscape²³.

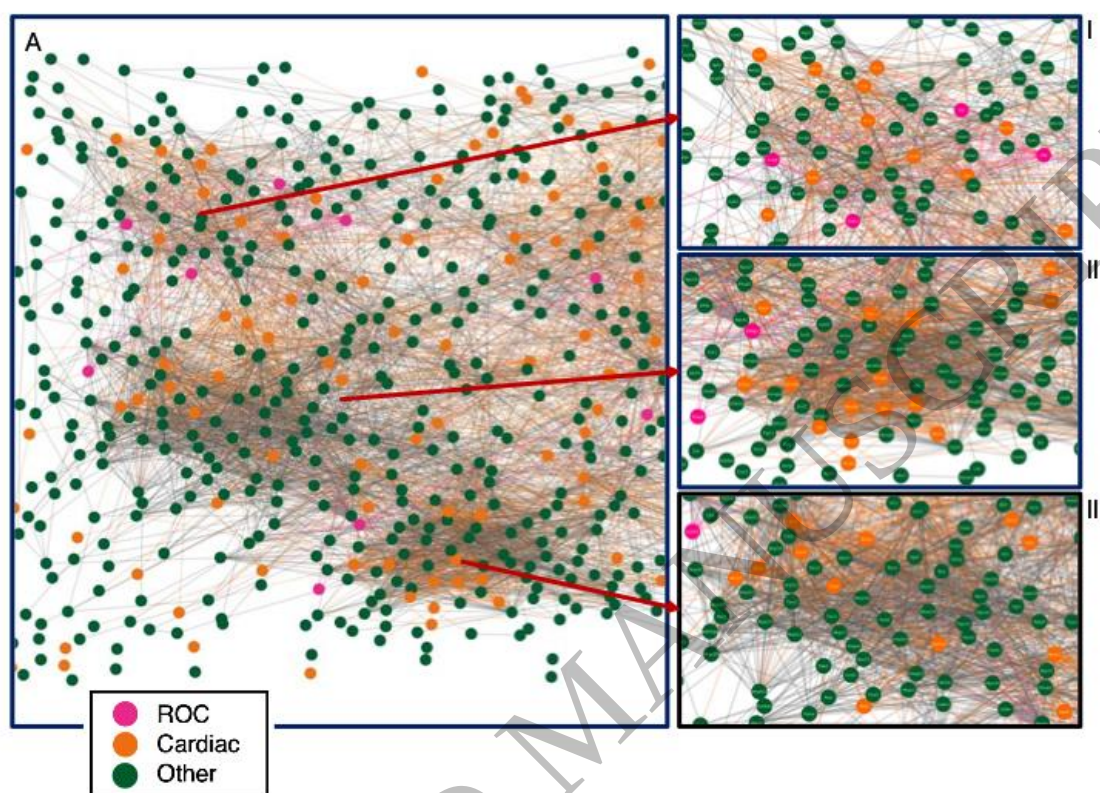


Figure 6. Protein-protein interaction network, where nodes correspond to Tau-associated proteins, while the edges represent potential protein-protein interactions, generated using Cytoscape/MCODE on STRING results. Orange-colored proteins are related to cardiac GO terms containing the following expressions: 'heart', 'muscle', 'contraction', 'I band'. Pink-colored proteins have been identified by the high AUC of ROC curves. **A) displays a large section of the full network, displaying proteins by interconnectedness. I-III) show close-ups of certain protein clusters, where closeness shows a higher degree of interaction on a proteomic level.**

Several of the Tau-related proteins which demonstrated an AUC > 0.85 were confirmed as hub genes by the MCODE analysis (Figure 6). Proteins related to cardiac GO terms containing the following expressions: 'heart', 'muscle', 'contraction', 'I band', and 'cardiac' were also identified and highlighted in the network plot. Furthermore, MCODE analysis allowed for the identification of three main protein clusters, or subnetworks (MCODE scores: 9.138, 5.419, and 3.231). These subnetworks highlight functionally closely related or interacting proteins within the network of Tau-associated proteins. To shine a light on the dynamic relationship among the members of the subnetworks, their pairwise correlation was calculated respective to each experimental condition (Co, AB, DB).

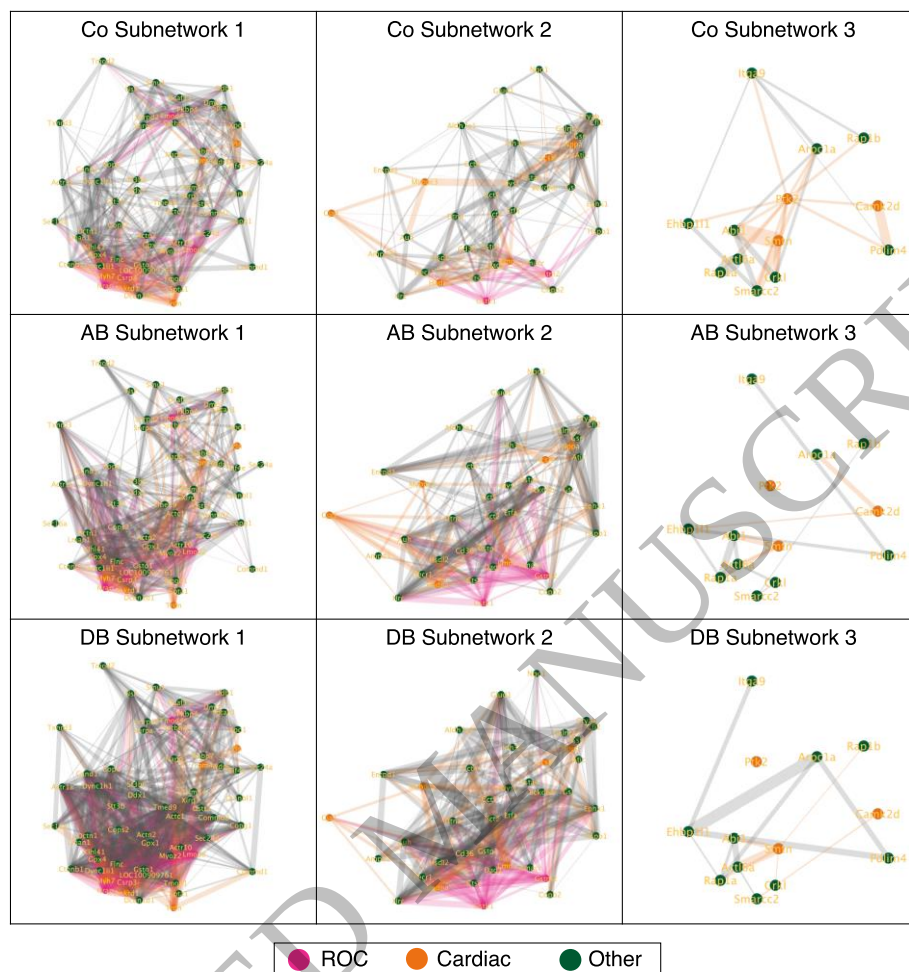


Figure 7. Network plots of proteins of closely interlinked subnetworks extracted from the main network identified by MCODE, based on potential protein-protein interactions of the STRING database (Subnetwork 1, MCODE Score 9.138; Subnetwork 2, MCODE Score 5.419; Subnetwork 3, MCODE Score 3.231). The edges represent the observed correlation coefficients in the different conditions (Co, AB, DB) among the proteins, which are depicted as nodes. The width of the edge (edge weight) increases with a higher correlation coefficient. Orange-colored proteins are related to cardiac GO terms containing the following expressions: 'heart', 'muscle', 'contraction', 'Tband', 'cardiac'. Pink-colored proteins had an AUC of more than 0.85 based on sensitivity and specificity calculation across various thresholds differentiating banded and non-banded rats. (Subnetwork 1, MCODE Score 9.138; Subnetwork 2, MCODE Score 5.419; Subnetwork 3, MCODE Score 3.231).

Finally, transcription factor analysis was performed using the ChEA3 algorithm, to find the most likely culprits, that may drive the changes in expression of Tau-associated proteins (Figure 8). The ten transcription factors with the highest significance were: MAX, REST, MYOD1, MYC, MYOG, ATF3, GATA1, MAFF, USF1, IRF1.

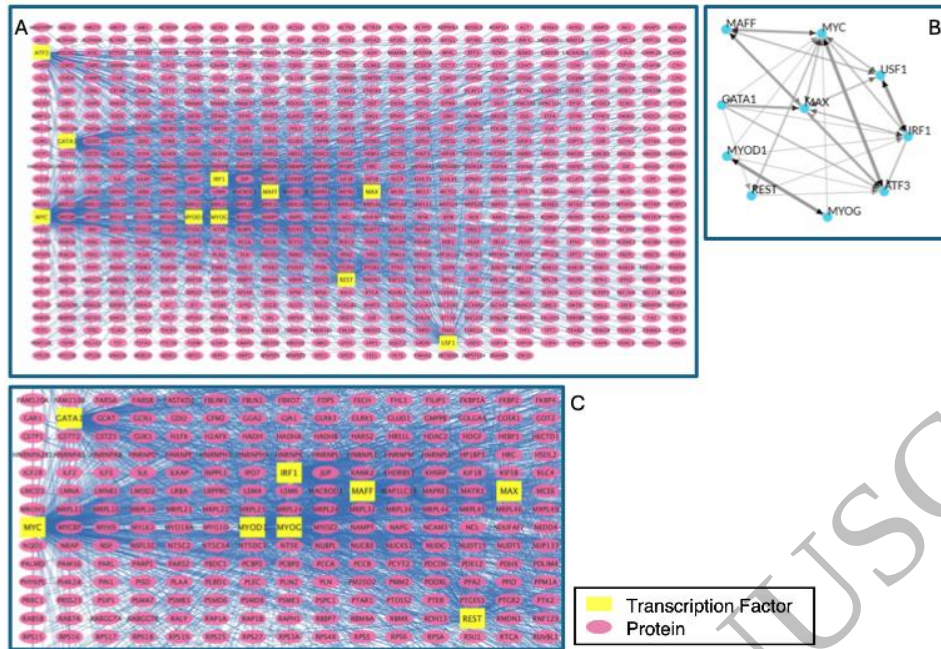


Figure 8. The ten most likely drivers (yellow) of Tau associated proteomic **remodelling** as identified by transcription factor analysis (ChEA3), and their target proteins (pink). A) displays the full web of transcription factors and their target proteins, to graphically represent the interconnectedness of the nodes. B) displays the transcription factors and the interactions amongst each. C) displays a close-up of selected nodes, for further emphasis on interconnectedness and clarity.

Discussion

Myocardial reverse remodelling remains difficult to quantify because structural and functional readouts do not directly reveal the molecular programmes that drive recovery or persistent dysfunction^{7,26}. In this study, we addressed that gap by linking standard echocardiographic and pressure-volume parameters to myocardial proteomic alterations in a rat model of pressure overload and unloading¹¹. This approach allowed us to ask not only which functional indices change during remodelling and reverse remodelling, but also which of them best reflect the underlying molecular state.

Functional assessment of reverse remodelling still lacks molecular anchoring

In clinical practice, reverse remodelling following pharmacological or mechanical pressure unloading is usually inferred from changes in chamber size, wall thickness, and global systolic performance^{7,27}, while advanced imaging, such as SPECT, can provide additional structural and functional detail. However, these measures do not consistently capture the degree of ongoing molecular recovery, and the criteria used to define reverse remodelling²⁸ vary substantially across studies. Thus, an important unmet need is to identify functional indices that are not only phenotypically informative, but also mechanistically linked to myocardial remodelling state.

Tau emerged as the most informative functional correlate of proteomic remodelling

Our findings indicate that chronic pressure overload drives a phenotype characterised by concentric remodelling, increased vascular load, and impaired diastolic relaxation. LASSO-based feature selection identified LV mass and Tau as the cardiac parameters most strongly linked to proteomic alterations associated with remodelling and reverse remodelling¹⁶. Among these, Tau showed the broadest set of protein associations, suggesting that active relaxation reflects a wider range of myocardial processes than morphology alone. This is biologically plausible, because efficient relaxation depends on coordinated calcium handling, ATP availability, cross-bridge detachment, cytoskeletal integrity, and adaptive gene expression²⁹⁻³². Disturbance in any of these domains can prolong isovolumic pressure decay. Tau therefore appears to function as an integrative readout of myocardial state, capturing not only the structural consequences of pressure overload but also the energetic and regulatory processes that support recovery after unloading.

Tau-linked proteins converge on sarcomeric, metabolic/redox, and proteostatic programmes

Compared with LV mass, Tau-associated proteins mapped to a broader set of Gene Ontology categories, including nucleic acid metabolic process, chromosome organization, and muscle tissue development. In addition, a set of 22 proteins, predominantly associated with Tau, was found to discriminate ongoing myocardial remodelling from control and debanded states with a ROCAUC of more than 0.85. Collectively, these proteins converged into three principal mechanistic domains. First, cytoskeletal and mechanotransductive components (e.g., TMOD4³³, PDLIM5³⁴, SORBS2³⁵, NRAP³⁶) highlight remodelling of sarcomeric architecture and force transmission^{35,37-39}. Second, metabolic and redox-related proteins (e.g., COQ9⁴⁰, GSTK1, DBT, PCCA, TST) implicate mitochondrial function and oxidative stress handling in impaired relaxation⁴¹⁻⁴⁷. Third, proteins involved in translation initiation and proteostasis (e.g., EIF4G1⁴⁸, PSME1, ANXA5) suggest altered protein turnover and stress-response signalling⁴⁹⁻⁵³. Efficient active relaxation depends on preserved energy supply, structural integrity, and coordinated protein synthesis; disruption across these axes provides a mechanistic framework linking proteomic remodelling to prolonged Tau during pressure overload.

Network and transcription factor analyses support coordinated remodelling biology

Protein-protein interaction analysis supported the view that Tau-associated proteins do not behave as isolated markers, but as components of coordinated remodelling modules (Figure 6-7). Subnetwork1 prominently features actin and myosin chain components, and proteins of the oxidative stress response (e.g., glutathione peroxidase). Subnetwork 2 is rich in proteins involved in the maintenance of muscle cell structure, from the nucleus to the extracellular matrix (Cryab, Dag1, Dtna, Dysf, and Lmna) and in metabolic proteins (Acsl6, Aldh7a1, Ampd3, Atic, Etfa, Glud1, Gsn, Gsr, and Nqo1). Finally, cytoskeletal organization, gene regulation, and signaling are the main functions of proteins in subnetwork 3 (Abi1, Actl6a, Arpc1a, Camk2d, Crkl, Ehbp1l1, Hist2h4, Itga9, Pdlim4, Ptk2, Rap1a, Rap1b, Smarcc2, Smtn). These clusters reinforce the idea that impaired relaxation emerges from concerted changes across multiple cellular systems rather than from dysfunction in a single pathway.

Ultimately, transcription factor enrichment analysis identified regulatory drivers that are presumably driving the observed Tau-associated proteomic remodelling (Figure 8). The highest-ranking factors included MAX and MYC, linked to growth regulation⁵⁴⁻⁵⁶; MYOD1 and MYOG, consistent with the reactivation of developmental and myogenic programmes^{57,58}; MAFF, associated

with oxidative stress responses⁵⁹; and ATF3 and IRF1⁶⁰, which are connected to adaptive and inflammatory stress signalling^{61,62}. These transcriptional regulators suggest coordinated control of metabolic adaptation, cytoskeletal restructuring, and stress-response pathways during pressure overload and unloading. Such transcriptional programmes provide a mechanistic bridge between haemodynamic stress and global proteomic reorganisation.

Clinical relevance and impact of proteomic observations

From a translational perspective, the present findings support active relaxation as a particularly informative physiological domain for tracking myocardial remodelling. Tau reflects the rate of isovolumic pressure decay and is relatively insensitive to preload^{63,64}, making it a more direct index of active relaxation than conventional echocardiographic surrogates such as E/e' or isovolumic relaxation time⁶⁵. In the present study, Tau was linked to reproducible proteomic remodelling states spanning cytoskeletal, metabolic/redox, and proteostatic pathways, suggesting that it captures not only haemodynamic dysfunction but also the underlying molecular state of the myocardium, including both disease severity and its potential reversibility. This is clinically relevant because the heterogeneity of heart failure likely contributes to the limited success of broadly applied therapies, and proteomic phenotyping may help identify biologically distinct remodelling states and actionable targets. In this context, Tau-associated proteins, including mitochondrial and redox-related factors and proteostatic components provide a rationale for exploring novel strategies to improve diastolic function. At the same time, Tau itself requires invasive pressure-volume analysis and therefore cannot be considered a routine clinical biomarker on the basis of the present data. The more immediate translational implication is that active relaxation, as a physiological domain, may be particularly informative for tracking molecular remodelling, and future studies should test whether non-invasive indices such as tissue Doppler e' velocity, E/e' ratio, diastolic strain rate, or advanced imaging-based relaxation measures capture similar biology during pressure overload and therapeutic unloading⁶⁵. If validated, such approaches could provide a bedside window into the evolving molecular state of the myocardium and support therapy monitoring in both experimental and clinical settings.

Limitations

It should be noted that while both pressure-volume and echocardiographic measurement techniques can be employed to different extents for an assessment of LV active relaxation, and thereby Tau values, in the scope of this study pressure-volume measurement techniques were employed. While we are aware of the wider accessibility of echocardiographic measurement techniques, pressure-volume measurement allowed us to gain a comprehensive insight into the intricacies of cardiac mechanics. It must be stressed, however, that due to the invasive nature of the procedure, pressure-volume measurement is not recommended for all patients in a routine clinical setting. Pressure-volume analysis, while the gold standard for Tau measurement, requires invasive catheterisation and carries procedural risk [ref. 63]. Whether non-invasive echocardiographic indices can serve as surrogates for Tau in this context, and with what accuracy, remains an open question that warrants dedicated validation studies^{65,66}. **Right ventricular structure and function were not assessed in this dataset and inherently thereby potential biventricular interactions**

1 unfortunately cannot be addressed. Finally, while pooling Co and DB animals for ROC analyses was
2 biologically justified, group-specific validation may be warranted in future studies.

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6 Author contributions

7 B.A. Barta, M. Ruppert, O. Schilling, and T. Radovits conceived and designed the study. BA. Barta, and
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12 manuscript and revising it critically for important intellectual content. B.A. Barta, T. Radovits and O.
13 Schilling coordinated the research project and finalized the manuscript for publication. T. Radovits, and O.
14 Schilling contributed equally to this work and share last authorship. All authors read and approved the final
15 manuscript.

16 Declaration of interest

17 None declared.

18 Data availability

19 Proteomic raw data is available from the MassIVE repository (dataset: MSV000089077,
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Legends

Graphical abstract. Illustration of data processing steps, including data collection and feature selection. AB = aortic banded, DB = debanded, EF = ejection fraction, SV = stroke volume, CO = cardiac output, FS = fractional shortening, RWT = relative wall thickness, MAP = mean arterial pressure, SW = stroke work, Ea = arterial elastance, EDPVR = end-diastolic pressure-volume relationship, PRSW = preload-related stroke work.

Table 1. Echocardiographic parameters in the experimental groups.

This table summarizes terminal echocardiographic measurements in the sham-operated control group (Co), the 6-week aortic banding group (AB6), the 12-week aortic banding group (AB12), and the debanded group (DB). Values are presented as mean $\pm$ SEM, with male and female animals pooled within each experimental group. Statistical comparisons were performed using one-way ANOVA followed by Tukey's post hoc test. Symbols indicate statistical significance as follows: * $p < 0.05$ vs Co; # $p < 0.05$ vs AB6; \$ $p < 0.05$ vs AB12. RWT, relative wall thickness; FS, fractional shortening; EF, ejection fraction; SV, stroke volume; CO, cardiac output; LV mass, left ventricular mass; SEM, standard error of the mean.

Table 2. Pressure-volume analysis parameters in the experimental groups.

This table summarizes terminal invasive pressure-volume analysis in the sham-operated control group (Co), the 6-week aortic banding group (AB6), the 12-week aortic banding group (AB12), and the debanded group (DB). Values are presented as mean $\pm$ SEM, with male and female animals pooled within each experimental group. Statistical comparisons were performed using one-way ANOVA followed by Tukey's post hoc test. Symbols indicate statistical significance as follows: * $p < 0.05$ vs Co; # $p < 0.05$ vs AB6; \$ $p < 0.05$ vs AB12. CO, cardiac output; EDPVR, end-diastolic pressure-volume relationship; PRSW, preload-recrutable stroke work; SW, stroke work; Ea, arterial elastance; MAP, mean arterial pressure; Tau, time constant of isovolumic relaxation; SEM, standard error of the mean.

Figure 1. Number of proteins associated with sensitive parameters of PV analysis and echocardiography, determined through univariate analysis (limma conducted on each parameter separately, p -value < 0.05) of all proteins and multivariate analysis of all proteins with feature selection by LASSO regression. For LASSO, we performed a subsequent limma analysis and define parameters with p -value < 0.05 as significantly associated with PV or echocardiographic parameters. Panel A illustrates that Tau emerges as the PV parameter associated with the highest number of proteins, and Panel B highlights LV mass as the most predominant echocardiographic parameter.

Figure 2. Gene Ontology (over-representation) enrichment analysis for Tau and LV mass-related proteins, identifying the molecular functions and cellular compartments of proteins associated with each parameter. The continuous color scale illustrates respective FDR adjusted p -values.

Figure 3. Enrichment of biological process for Tau-related proteins. To provide a better overview, terms were automatically grouped based on semantic similarity.

Figure 4. Enrichment of biological processes for LV mass-related proteins. To provide a better overview, terms were automatically grouped based on semantic similarity.

Figure 5. ROC curves of proteins associated to Tau with an AUC of more than 0.85, based on the calculation of sensitivity and specificity across different thresholds, differentiating active myocardial remodelling (AB) from non-banded (CO, DB) rat proteomes.

Figure 6. Protein-protein interaction network, where nodes correspond to Tau-associated proteins, while the edges represent potential protein-protein interactions, generated using Cytoscape/MCODE on STRING results. Orange-colored proteins are related to cardiac GO terms containing the following expressions: 'heart', 'muscle', 'contraction', 'I band'. Pink-colored proteins have been identified by the high AUC of ROC curves. A) displays a large section of the full network, displaying proteins by interconnectedness. I-III) show close-ups of certain protein clusters, where closeness shows a higher degree of interaction on a proteomic level.

Figure 7. Network plots of proteins of closely interlinked subnetworks extracted from the main network identified by MCODE, based on potential protein-protein interactions of the STRING database (Subnetwork 1, MCODE Score 9.138; Subnetwork 2, MCODE Score 5.419; Subnetwork 3, MCODE Score 3.231). The edges represent the observed correlation coefficients in the different conditions (Co, AB, DB) among the proteins, which are depicted as nodes. The width of the edge (edge weight) increases with a higher correlation coefficient. Orange-colored proteins are related to cardiac GO terms containing the following expressions: 'heart', 'muscle', 'contraction', 'I band', 'cardiac'. Pink-colored proteins had an AUC of more than 0.85 based on sensitivity and specificity calculation across various thresholds differentiating banded and non-banded rats. (Subnetwork 1, MCODE Score 9.138; Subnetwork 2, MCODE Score 5.419; Subnetwork 3, MCODE Score 3.231).

Figure 8. The ten most likely drivers (yellow) of Tau associated proteomic remodelling as identified by transcription factor analysis (ChEA3), and their target proteins (pink). A) displays the full web of transcription factors and their target proteins, to graphically represent the interconnectedness of the nodes. B) displays the transcription factors and the interactions amongst each. C) displays a close-up of selected nodes, for further emphasis on interconnectedness and clarity